

House Price Prediction using Model Validation, Overfitting Control & Hyperparameter Tuning

Introduction

This project evaluates Linear Regression, Decision Tree Regressor, and a Tuned Decision Tree Regressor on the California Housing dataset using train-test split, cross-validation, and GridSearchCV.

Objective

Perform model validation, detect overfitting, tune hyperparameters, and compare regression models.

Dataset

California Housing dataset from scikit-learn.

Technologies

Python, NumPy, Pandas, Matplotlib, scikit-learn, Joblib, Jupyter Notebook.

Methodology

Loaded dataset, performed EDA, train-test split, trained baseline models, applied 5-fold cross-validation, tuned Decision Tree using GridSearchCV, evaluated with RMSE and R^2 .

Results

Linear Regression: RMSE=0.745581, R^2 =0.575788

Decision Tree: RMSE=0.709301, R^2 =0.616068

Tuned Decision Tree: RMSE=0.639065, R^2 =0.688338

Conclusion

The Tuned Decision Tree achieved the lowest RMSE and highest R^2 , demonstrating that hyperparameter tuning improved model performance and reduced overfitting compared with the baseline models.

Future Work

Experiment with Random Forest, Gradient Boosting, XGBoost, feature engineering, and deployment.